BEHARA GAMYA SRI
2211CS010035 (S1)

IBM HR Analytics Employee Attrition & Performance Analysis

INTRODUCTION

- The project focuses on analyzing employee attrition within an organization using the IBM HR Analytics dataset. Employee attrition (turnover) refers to employees leaving the company, which can lead to increased costs, disruption in workflows, and loss of talent. The goal is to identify patterns and factors contributing to attrition and provide actionable insights to reduce employee turnover and improve workforce management.

1. DATASET OVERVIEW

The IBM HR Analytics dataset is a fictional dataset created by IBM data scientists to help uncover factors contributing to employee attrition. It allows exploration of important questions such as the impact of distance from home by job role, or the relationship between average monthly income, education, and attrition.

- **Total Records:** 1,470 employees
- **Target Variable:** Attrition (Yes/No)

1.1 SOURCE

The dataset is a fictional HR dataset created by IBM data scientists and available on Kaggle (IBM HR Analytics Employee Attrition Dataset). It provides detailed employee information to analyze attrition patterns and workforce trends effectively.

1.2 COLUMNS/ FEATURES

The dataset contains 35 columns capturing employee demographics, work experience, performance, and organizational attributes. Key columns used in this analysis include:

- Age – Employee age
- Attrition – Target variable (Yes/No)
- BusinessTravel – Travel frequency for work
- DailyRate, HourlyRate, MonthlyIncome, MonthlyRate – Compensation details
- Department – Employee department
- DistanceFromHome – Distance between home and workplace
- Education, EducationField – Education level and specialization
- EmployeeCount, EmployeeNumber, Over18, StandardHours – Organizational identifiers
- EnvironmentSatisfaction, JobSatisfaction, RelationshipSatisfaction – Employee satisfaction metrics
- JobInvolvement, JobLevel, JobRole – Role and job-related attributes
- MaritalStatus, Gender – Personal attributes
- NumCompaniesWorked, TotalWorkingYears, YearsAtCompany – Career and experience history
- YearsInCurrRole, YearsSinceLastPromotion, YearsWithCurrManager – Career and experience history
- OverTime – Whether the employee works overtime
- PercentSalaryHike, PerformanceRating, StockOptionLevel – Performance, benefits
- TrainingTimesLastYear, WorkLifeBalance – training metrics

1.3 DATA QUALITY

- Well-structured
- No missing values in key columns used for analysis.
- Categorical variables (properly labeled, while numerical columns and in correct datatypes)
- The dataset allows for meaningful aggregation, transformation, and visualization to analyze attrition trends effectively.

1.4 KEY STATISTICS

- Total Records: 1,470 employees
- Department Distribution: 3 main departments – Sales, R & D, Human Resources
- Income & Age Statistics:
  - Monthly Income ranges from $1,000 to $20,000
  - Age ranges from 18 to 60 years
- Overtime: ~30% employees work overtime

2. OPERATIONS PERFORMED

2.1 DATA CLEANING AND EXPLORATION

- Checked dataset structure and schema using Spark.
- No missing/null values observed in key columns used for analysis.
- Inspected first few rows (head) and last few rows (tail) for initial understanding.
- Converted categorical Attrition column to numeric (Attrition_numeric) for aggregation.
- Created derived columns:
    - AgeGroup (<30, 30-39, 40-49, 50+)
    - IncomeBand (Low, Medium, High, Very High)

2.2 DESCRIPTIVE ANALYSIS

- Calculated average attrition rate by Department and visualized using bar plots.
- Compared Gender-based attrition using pie charts.
- Analyzed attrition impact of 'OverTime' using bar charts.
- Examined attrition trends by 'IncomeBand' with bar plots.
- Studied effect of 'YearsSinceLastPromotion' on attrition using line plots.
- Explored attrition across 'AgeGroup' and 'EducationField' with bar charts.
- Assessed influence of 'JobSatisfaction' levels on attrition with bar charts.

2.3 RELATIONSHIPS ANALYSIS

- Department vs. Attrition – Identified departments with the highest attrition percentages.
- Gender × Attrition – Compared attrition trends across male and female employees.
- OverTime × Attrition – Assessed the impact of extra working hours on employee turnover.
- IncomeBand × Attrition – Studied how salary levels relate to attrition rates.
- Promotion × Attrition – Explored the relationship between promotion gaps and attrition.
- AgeGroup × Attrition – Analyzed how age affects likelihood of leaving
- JobSatisfaction × Attrition – Examined how satisfaction levels relate to attrition.
- JobRole × OverTime × Attrition – Combined role and overtime to identify which job roles, when coupled with overtime, have higher attrition.
- Multiple Factors × Attrition – Combined key attributes (Department, OverTime, Income, Promotion Gap, Age, JobSatisfaction) to analyze their collective impact on attrition.

3. KEY INSIGHTS/ FINDINGS

3.1 WORKFORCE DEMOGRAPHICS

- Younger employees (<30 years) show higher attrition.
- Mid-career employees (30–39 years) have moderate attrition
- employees above 40 show lower attrition.
- Job satisfaction affects retention across all age groups.

3.2 DEPARTMENTAL INSIGHTS

- Sales department exhibits the highest attrition.
- Other departments have moderate or lower attrition.

3.3 WORK PATTERNS

- Employees working OverTime are more likely to leave.
- Combining JobRole and OverTime identifies high-risk groups.

3.4 COMPENSATION & PROMOTION

- Lower-paid employees have higher attrition.
- Employees without recent promotions tend to leave.

3.5 JOB SATISFACTION & ENGAGEMENT

- Low job satisfaction strongly correlates with attrition.
- Monitoring satisfaction is especially important for younger employees.

3.6 OVERALL INSIGHTS

- Attrition is multifactorial; no single factor explains all departures.
- Visualizations help quickly identify priority areas for HR action.
- Insights guide retention programs, work-life balance initiatives, compensation & promotion strategies, and engagement plans.

4. RECOMMENDATIONS & FUTURE ENHANCEMENTS

4.1 Talent Development & Retention

- Focus retention programs on high-risk groups (Sales, younger employees, low job satisfaction).
- Monitor career growth and promotions to improve retention.

4.2 Compensation & Workload

- Ensure fair pay and timely promotions.
- Reduce excessive overtime and improve work-life balance.

4.3 Department & Role Management

- Address high attrition in Sales; monitor other departments for stability.
- Consider role-specific interventions for JobRole and EducationField.

4.4 Future Analytics Opportunities

- Build predictive models to proactively identify at-risk employees.
- Integrate additional data (performance, engagement) for deeper insights.
- Create dashboards for real-time attrition monitoring and decision-making.

5. CONCLUSION

The analysis of IBM HR Analytics data revealed key factors driving employee attrition, including age, Sales department, overtime, low pay, and limited promotions. Insights from this study can help HR design targeted retention programs, improve compensation and career growth strategies, and enhance employee engagement. Future predictive models and dashboards can further support proactive workforce management and reduce turnover.